



Certificate / Certificat Zertifikat / 合格証

Parker 080228 P0001 C001.11

exida hereby confirms that the:

Solenoid Valves types F, V, X and N

**Parker Hannifin Manufacturing SRL
Corsico (MI)
ITALY**

The manufacturer
may use the mark:



Revision 5.0 February 19, 2020

Surveillance Audit Due
April 1, 2023

Has been assessed per the relevant requirements of:

IEC 61508 : 2010 Parts 1 and 2

and meets requirements providing a level of integrity to:

Systematic Integrity: SC 3 (SIL 3 Capable)

Random Integrity: Type A Device

**PFD_{AVG} and Architecture Constraints
must be verified for each application**

Safety Function:

The solenoid valve will move to the designed safe position when de-energized / energized within the specified safety time.

Application Restrictions:

The unit must be properly designed into a Safety Instrumented Function per the requirements in the Installation, Operations and Maintenance and Safety Manuals.



ISO/IEC 17065
PRODUCT CERTIFICATION BODY
#1004



Evaluating Assessor

Certifying Assessor

Systematic Capability: SC 3 (SIL 3 Capable)**Random Capability: Type A Device**

**PFH/PFD_{avg} and Architecture Constraints
must be verified for each application**

**Solenoid Valves types
F, v, X and N**

Systematic Capability:

The product has met manufacturer design process requirements of Safety Integrity Level (SIL) 3. These are intended to achieve sufficient integrity against systematic errors of design by the manufacturer.

A Safety Instrumented Function (SIF) designed with these products must not be used at a SIL level higher than stated.

Random Capability:

The SIL limit imposed by the Architectural Constraints must be met for each element.

**IEC 61508 Failure Rates in FIT* with partial valve stroke testing (PVST)
Profile 3, Solenoid Valve types F, V, X and N:**

Valve Body	Insulation	Power	λ_{Safe}	λ_{DD}	λ_{DU}
U131F5695	H	6W, 8W	107	61	5
U121V5596	H	6W, 8W	107	61	5
U033X5152	H	3W, 6W, 8W	76	137	31
131X1131	H	6W, 8W	76	137	31
U133X5192	H	6W, 8W	76	137	31
341 N3197	H	1.6 W	57	507	147

All failure rates are given in FIT=10⁻⁹/h

For a complete overview of all certified valves, please see the Certificate Annex V5 R0.

SIL Verification:

The Safety Integrity Level (SIL) of an entire Safety Instrumented Function (SIF) must be verified via a calculation of PFD_{AVG} considering redundant architectures, proof test interval, proof test effectiveness, any automatic diagnostics, average repair time and the specific failure rates of all products included in the SIF. Each subsystem must be checked to assure compliance with minimum hardware fault tolerance (HFT) requirements

The following documents are mandatory parts of this certificate:

Assessment report: Parker Hannifin 0802-28-C R004 V5R0

Safety Manual: See annex 1 in assessment report.



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